

Karthik Rao

| Senior Product Leader – Data, IoT & Decision Intelligence Platforms

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SKILLS

Product Management | Product Strategy | Vision & Roadmap | GTM | Prioritization | Competitive Analysis | Customer Insights | Product Development Lifecycle

Program & Project Management | Stakeholder Management | Cross-functional Leadership (Eng, DS, Design, Business) | Budgeting | Risk Management | Process Improvement

Data & Analytics | SQL | Python - Data Analysis | Data Modeling | Power BI | Tableau, Excel | A/B Testing | EDA | ETL | RFM Analysis | Probability & Statistics | Metrics Design (OKR, KPI)

Machine Learning & AI | Regression | Classification | Logistics | Clustering | Tree based models | Time Series | Recommendation Ranking | NLP | Computer Vision | Gen AI | LLMs | RAG | Agents | AI Integration & governance

MLOps and /Tools: MLFlow | Docker | Cloud AWS Sagemaker | REST APIs | CI/CD | System Design | GitHub | Agile/Scrum | JIRA, Confluence | SDLC | IoT

EXPERIENCE

Independent Freelance Consultant, *Self Employed*

Nov 2020 - Present

- Built AI-driven decision intelligence platform for retail investors, using ML-based anomaly detection and rule-based analytics to deliver real-time ranked insights, improving decision speed for 100+ users

Product Consultant, *Sterna Security Devices Pvt Ltd, India*

Feb 2024 - Mar 2025

Owned and scaled a cloud-connected IoT security platform to **5K+ devices**, enabling remote device management and reducing operational costs by **40% through automation and fleet intelligence capabilities**.

Tech: AWS IoT Core, Python, SQL, REST APIs, Docker, CI/CD

- Launched and scaled the IoT platform to drive **\$2M+ ARR within Year 1**, enabling international expansion and enterprise adoption across multiple markets
- Built a cloud-based IoT operations and telemetry layer enabling real-time fleet monitoring, remote provisioning, and automated device lifecycle management at scale
- Drove enterprise compliance initiatives across **ISO 27001, GDPR, and CE frameworks**, embedding security-by-design principles into IoT architecture and enabling **\$500K+ EU revenue expansion opportunity**

Digital Product Manager, *Buckman Inc. USA*

Jul 2020 - Oct 2020

Vision: Owned end-to-end IoT device management and intelligence platform transforming fragmented legacy systems into a unified, real-time industrial IoT ecosystem across chemical manufacturing sites.

Tech: Azure IoT Core, Python, SQL, MQTT, REST APIs, Real-time Streaming Analytics, Docker

- Built and scaled an enterprise IoT platform modernizing legacy systems, growing from **500 to 5,000+ connected devices across 15 manufacturing sites**, improving operational visibility and reducing costs by **35% through unified real-time telemetry architecture**
- Launched a cloud-based IoT provisioning and lifecycle management platform for **5,000+ devices**, enabling secure onboarding and automated lifecycle state management, reducing provisioning time from **2 weeks → 2 days (85% improvement)**
- Built real-time IoT analytics platform for industrial telemetry enabling hazard detection and operational insights, improving decision speed and contributing to **\$1M+ in avoided safety incidents and downtime reduction**

Technical Product Manager, *Dell Inc. USA*

Dec 2015 - Nov 2019

Vision: Built enterprise Decision Intelligence, KPI governance, and engineering productivity platforms to unify fragmented reporting and execution systems into real-time, trusted enterprise decision-making.

Tech: SQL, Python, REST APIs, Power BI, Klipfolio, Power Platform, Jenkins, Office 365

- Built and scaled a Decision Intelligence Platform from **0 → 500+ users across 5+ business functions**, reducing reporting latency by **70%** through a unified real-time analytics layer replacing fragmented dashboards.

- Defined product strategy and secured **\$1.5M executive funding** for enterprise analytics modernization, driving cross-functional adoption across Finance, Engineering, Sales, and Operations within 6 months
- Established an enterprise **KPI Governance & Center of Excellence**, standardizing metrics across 10 departments and improving executive reporting trust by eliminating KPI inconsistencies
- Owned enterprise **engineering productivity and analytics ecosystem strategy**, aligning tooling investments and platform decisions with business outcomes, improving adoption consistency and enabling measurable ROI visibility
- Delivered **\$2M–\$3M in business value** through analytics-driven optimization, improved decision velocity, and enterprise-wide adoption of unified decision systems.

Software Engineer, Dell Inc.

Dec 2008 - Nov 2015

Vision: Transformed manual QA processes into an automated CI-driven quality intelligence system improving release velocity, reliability, and engineering efficiency at scale.

Tech Stack: Java, Selenium, Jenkins, CI/CD pipelines, TestNG, Python, SQL

- Built an enterprise CI-driven test automation platform adopted across **50+ engineering teams**, reducing release cycles from **4 weeks to 3 days (95% improvement)** and accelerating delivery velocity
- Eliminated manual QA bottlenecks by introducing scalable automated regression pipelines, reducing QA effort by **90% (200 hrs/sprint → 20 hrs/sprint)** and increasing test coverage to **80% across the organization**

AI PRODUCT LEADERSHIP PORTFOLIO

Real-Time Retail Recommendation System | *Repo:* github.com/Karthik-pmlead

- Built an event-driven, real-time recommendation platform (Amazon/Netflix-style) that converts behavioral and session signals into context-aware personalization, improving simulated engagement (CTR/session duration) by 25%
- Designed a hybrid ranking and experimentation framework combining collaborative filtering, session-based modeling, and real-time A/B testing pipelines, improving recommendation relevance (NDCG@10) by 20% while ensuring explainability and user trust

AI Decision Intelligence Platform | *Repo:* github.com/Karthik-pmlead

- Built an agentic decision intelligence platform that transforms 100K+ multi-source business, operational, and customer signals into real-time, explainable insights and recommendations, reducing manual analysis effort by 60% and accelerating decision turnaround by 40%
- Designed an event-driven, multi-agent orchestration system integrating anomaly detection, sentiment analysis, trend intelligence, and recommendation engines with streaming data pipelines and explainability layers for transparent decision support

Operational Intelligence Platform | *Repo:* github.com/Karthik-pmlead

- Built a multilingual sentiment intelligence platform converting 100K+ customer feedback events into actionable insights, achieving 94% sentiment classification accuracy on benchmark data
- Designed an AI-driven observability system with streaming dashboards and sentiment analytics, improving simulated response times by 50% via proactive alerts and decision support.

EDUCATION

- **MBA, Management of Technology**, Georgia Institute of Technology, USA, 2013
- **M.S Electrical Engineering** (Major: Computer Science) Wichita State University, USA, 2008
- **B.E Electrical Engineering**, Anna University, India, 2005

CERTIFICATIONS

- **Scaler** - Data Science and Machine Learning, 2026
- **Project Management** Course, 2025
- **Product School** - Product Management Certificate, 2020
- **Udacity** - Data Analyst, Data Engineering, AI Product Manager, AWS Cloud Architect, 2020-22